

# Dementia Risk Prediction with Gaussian Process Classification

## DATASCI 224 Final Project SP2026

Jazmine Haratani

## Introduction

Dementia affects tens of millions of adults worldwide, with Alzheimer's Disease standing as its most prevalent and devastating form. As global populations age, understanding the biological and cognitive predictors of dementia progression has become one of the most urgent challenges in neuroscience and public health. This analysis draws on data from the Open Access Series of Imaging Studies (OASIS), a publicly available collection of neuroimaging datasets compiled to support research and open scientific inquiry.<sup>1</sup> Specifically, the dataset used here is the OASIS Longitudinal Study of Nondemented and Demented Older Adults, comprising MRI data from 150 subjects between the ages of 60 and 96.

Beyond neuroimaging, the dataset captures an important set of participant characteristics: demographic variables including age, sex, and handedness; socioeconomic and educational background (years of education and SES); and a battery of clinically meaningful measures — the Mini-Mental State Examination (MMSE), normalized whole brain volume (nWBV), Atlas Scaling Factor (ASF), and estimated total intracranial volume (eTIV). The primary outcome of interest is the Clinical Dementia Rating (CDR), a widely used clinical staging tool for characterizing dementia severity. This project focuses on more interpretable predictors of CDR — namely age, education, socioeconomic status (SES), sex (M/F), MMSE, and nWBV — to illuminate the cognitive and structural factors most closely associated with dementia progression. We hypothesized that males with a higher age and lower MMSE score, nWBV, and number of years of education would show a higher probability of dementia.

## Methods

This project examines whether age, sex, education, SES, MMSE scores, and MRI-derived brain volume (nWBV) predict dementia status over time in older adults, using Gaussian Process Classification to predict probability of dementia given these features, as well as the confidence in that result. The MMSE is a validated test of cognitive function consisting of eleven questions scored from 0 (worst) to 30 (best).<sup>2</sup> The nWBV is expressed as a percent of all voxels in the atlas-masked image labeled as gray or white matter by automated tissue segmentation.<sup>3</sup> The CDR rates subjects across six domains — memory, orientation, judgment

and problem solving, community affairs, home and hobbies, and personal care — with a global CDR of 0 indicating no dementia and scores of 0.5, 1, 2, and 3 representing very mild, mild, moderate, and severe dementia, respectively.

## Gaussian Process Classification

Gaussian Process Classification (GPC) models uncertainty over functions, then uses a sigmoid link to map those functions to class probabilities. We begin by specifying a Gaussian Process over the latent function  $f(x)$ , where  $x$  is a patient's feature vector and  $f(x)$  is the corresponding latent risk score — a continuous, unobserved quantity reflecting underlying dementia burden. Formally:

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

Where:

$m(x)$  = mean function, typically set to 0

$k(x, x')$  = kernel (covariance) function

The kernel encodes the prior belief that patients with similar feature vectors should have similar latent risk scores. Specifically, it measures the correlation between  $f(x)$  and  $f(x')$ : when  $x = x'$ ,  $k$  equals 1, and as inputs diverge,  $k$  approaches 0. The Radial Basis Function (RBF) kernel is the standard choice for this:

$$k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right)$$

Where the length-scale  $\ell$  controls how quickly correlation decays with distance in the input space.

Once the GP is fit to the dataset, the learned kernel length-scales  $\ell$  serve as a direct measure of feature relevance — a small  $\ell$  indicates the model is sensitive to variation in that variable (high relevance), while a large  $\ell$  indicates the model is largely insensitive to it (low relevance). We use the Automatic Relevance Determination (ARD) variant of the RBF kernel, which assigns an independent length-scale parameter  $\ell_d$  to each feature, making interpretability more direct:

$$k(x, x') = \exp\left(-\sum_{d=1}^D \frac{(x_d - x'_d)^2}{2\ell_d^2}\right) \quad \text{and} \quad K_{ij} = k(x_i, x_j)$$

A large  $\ell_d$  implies that the function barely changes as feature  $d$  varies, suggesting that feature contributes little to the prediction.

Evaluating the kernel over all pairs of training patients produces an  $(n \times n)$  covariance matrix  $K$ , which encodes the prior belief structure of the model. The GP prior over the training latent scores is then:

$$f \sim \mathcal{N}(\mathbf{0}, K)$$

The GP places a prior over the latent function  $f(x)$ , which is subsequently passed through a sigmoid function to produce a probability of dementia. However, because the outcome is binary, the likelihood is Bernoulli rather than Gaussian. This creates a fundamental challenge: the posterior is proportional to a Gaussian prior multiplied by a Bernoulli likelihood, which does not yield a closed-form Gaussian. We therefore apply the Laplace approximation, which approximates the true non-Gaussian posterior with the Gaussian that best fits at its peak.

The relevant quantities are defined as follows:

Likelihood:

$$p(y_i | f_i) = \sigma(f_i)^{y_i} [1 - \sigma(f_i)]^{1-y_i}$$

Sigmoid (logistic) function:

$$\sigma(f) = \frac{1}{1 + e^{-f}}$$

Posterior:

$$p(f | y) \propto p(y | f) p(f)$$

The Laplace approximation proceeds in two steps.

### 1 – Maximize the log posterior using Newton's method:

$$\log p(f | y) = \sum_i \log p(y_i | f_i) - \frac{1}{2} f^T K^{-1} f - \frac{1}{2} \log |K|$$

$$W = -\nabla \nabla \log p(y | f) = \text{diag}(\pi_i(1 - \pi_i))$$

Where  $\pi_i = \sigma(f_i)$  is the current predicted probability for patient  $i$  — that is, the probability of the observed CDR label given the current latent risk score estimate. Newton's method iteratively refines the latent scores by balancing two competing forces: fitting the observed labels and remaining consistent with the prior. The Newton update at each iteration is:

$$f^{\text{new}} = (K^{-1} + W)^{-1} W \tilde{f}$$

Where  $\tilde{f} = f + W^{-1} \nabla \log p(y | f)$  is the adjusted target. This iterates until convergence to the MAP estimate  $\hat{f}$ .

### 2 – Fit a Gaussian at the peak mode:

Once  $\hat{f}$  is identified, the Laplace approximation fits a Gaussian centered at the mode:

$$q(f | y) = \mathcal{N}(\hat{f}, \Sigma)$$

Where the covariance is the negative inverse Hessian of the log posterior evaluated at the mode:

$$\Sigma = (K^{-1} + W)^{-1}$$

Here,  $K$  encodes prior smoothness and  $W$  encodes the curvature of the likelihood at each point. Where training data is dense and labels are consistent (high  $\pi_i(1 - \pi_i)$ ), the likelihood dominates; where data is sparse, the prior dominates.

In summary, Newton's method identifies the most plausible set of latent risk scores given the observed labels, and the Laplace approximation characterizes the uncertainty around that estimate by fitting a Gaussian to the posterior at its peak.

**Predictions.** With the approximate posterior in hand, we ask: given everything learned from training, what is the latent risk score for a new patient  $x_*$ ? The predictive distribution over  $f_*$  is given by:

$$\mu_* = \mathbf{k}_*^T K^{-1} \hat{\mathbf{f}}$$

$$\sigma_*^2 = k_{**} - \mathbf{k}_*^T (K + W^{-1})^{-1} \mathbf{k}_*$$

Where  $\mathbf{k}_* = [k(x_*, x_1), \dots, k(x_*, x_n)]^T$  is the vector of kernel evaluations between the new patient and all training patients, and  $k_{**} = k(x_*, x_*)$ . The predictive mean  $\mu_*$  is a weighted sum of training latent values, with nearby patients (high  $k(x_*, x_i)$ ) contributing more. The predictive variance  $\sigma_*^2$  begins at the prior variance  $k_{**}$  and is reduced proportionally to how well the training data constrains the prediction — it remains large in sparse regions of the input space and shrinks near dense, consistently-labeled training data.

The final class probability is obtained by passing the predictive distribution through the sigmoid, well-approximated by:

$$p(y_* = 1 \mid x_*, \mathbf{y}) \approx \sigma \left( \frac{\mu_*}{\sqrt{1 + \frac{\pi \sigma_*^2}{8}}} \right)$$

When  $\sigma_*^2$  is large, the predicted probability is pulled toward 0.5, reflecting genuine uncertainty. When  $\sigma_*^2 \approx 0$ , the prediction reduces to  $\sigma(\mu_*)$ , recovering behavior similar to standard logistic regression.

**Hyperparameter learning.** The kernel hyperparameters ( $\ell_d$  for each feature) are learned by maximizing the log marginal likelihood:

$$\log p(\mathbf{y} \mid X, \boldsymbol{\theta}) = \log \int p(\mathbf{y} \mid f) p(f \mid X, \boldsymbol{\theta}) df$$

Under the Laplace approximation this integral becomes tractable. Gradient ascent via L-BFGS is used, with `n_restarts_optimizer` controlling the number of random restarts to avoid

local optima. This is the step at which the ARD length-scales are learned — the optimization simultaneously finds  $\hat{f}$  and the  $\ell_d$  values that best explain which features predict CDR status.

**Binary outcome.** Applying GPC to a binary outcome requires justification given the ordinal structure of the original CDR scale. A continuous outcome was not used, as predictions falling outside the clinically defined CDR range (0, 0.5, 1) would not be meaningfully interpretable. Treating CDR as an unordered three-class outcome was similarly unappealing, as this would ignore the inherent ordering and treat the difference between 0 and 0.5 as equivalent to the difference between 0.5 and 1, which does not reflect clinical reality. The most defensible threshold is the presence of any cognitive impairment — CDR = 0 versus CDR > 0 — making a binary outcome both clinically justified and directly aligned with the research question.

## Analysis and Application of GPC

We began by loading our OASIS-2 dataset directly into a dataframe, dropping any rows with missing values for the primary predictors or outcome variables. GPC cannot handle NaN values, and socioeconomic status as well as MMSE contained a few missing values in this dataset. We then subset to our five predictors of interest - Age, Education, SES, MMSE, sex, and nWBV - and copied them into a variable X, where we then scaled all to zero mean and unit variance. This step was vital in the process due to sensitivities GPC has, specifically with the kernel measures, and the features with larger numeric ranges potentially dominating the distance calculation between feature vectors. We then defined our Automatic Relevance Determination (ARD) Radial Basis Function (RBF) kernel, assigning one length-scale per feature, all set to one as starting values. The length scales were our interpretable feature importance at the end of our analysis, and eventually learned true values during the fitting of the training data.

```
In [17]: from IPython.display import display, HTML
import pandas as pd
import numpy as np
from sklearn.gaussian_process import GaussianProcessClassifier
from sklearn.gaussian_process.kernels import RBF, Matern, RationalQuadratic
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import StratifiedKFold, cross_val_score
from sklearn.metrics import classification_report, roc_auc_score

df = pd.read_excel('/home/jharatani/DATASCI224_Final_Project/oasis.xlsx') # Here we

# 1. Prepare features – binary outcome: CDR > 0
df = df.dropna(subset=["Age", "EDUC", "SES", "nWBV", "CDR"]) # Now we drop all rows

X = df[["Age", "EDUC", "SES", "M/F", "nWBV"]].copy() # Subset to predictor variable
X["M/F"] = (X["M/F"] == "M").astype(int) # encode sex as an integer
y = (df["CDR"] > 0).astype(int) # y is our dataframe where CDR > 0 is saved as an i
```

```
In [ ]: # 2. Scale – critical for GPC; kernel length-scales assume comparable magnitudes
scaler = StandardScaler() # Pre-processing - define our scaler function
X_scaled = scaler.fit_transform(X) # Scale features to zero mean and unit variance

# 3. Define kernel – ARD (Automatic Relevance Determination) variant
# One length-scale per feature: reveals variable importance
kernel = RBF(length_scale=np.ones(X_scaled.shape[1])) # Initialize ARD RBF kernel w
```

We instantiated our GaussianProcessClassifier() with the ARD kernel we defined, five optimizer restarts, and a fixed random state. The five optimizer restarts increased the chance of finding the global optimum for the length-scales. Following our model definition, we used cross-validation to stratify our dataset and produce an estimate of how well the model generalized, as interpreted by our AUC result. We did not use a train/test split with this dataset since the dataset is so small. This would have resulted in an inaccurate or noisy, unstable AUC result. Additionally, the stratification preserved the CDR class ratio that was observed in the full dataset, which may have been more important due to the higher number of cases of CDR > 0 in this case. Once we had the mean AUC and standard deviation over the 5 folds, we moved to our final model fit.

```
In [9]: # 4. Fit GPC – pass ARD kernel, allow 5 random restarts to avoid local optima in hy
gpc = GaussianProcessClassifier(kernel=kernel, n_restarts_optimizer=5,
                                random_state=42) # random_state for reproducibili
```

```
In [10]: # 5. Cross-validate using stratified k-fold to preserve class balance across folds
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Score each fold using AUC-ROC – appropriate for binary classification with potent
auc_scores = cross_val_score(gpc, X_scaled, y, cv=cv, scoring="roc_auc")

# Report mean and standard deviation of AUC across folds
print(f"Mean AUC: {auc_scores.mean():.3f} ± {auc_scores.std():.3f}")
```

Mean AUC: 0.775 ± 0.046

```
In [18]: performance_table = pd.DataFrame({
    "Metric": ["Mean AUC-ROC", "SD across folds", "Cross-validation"],
    "Value": [f"{auc_scores.mean():.3f}", f"± {auc_scores.std():.3f}", "5-fold stra
    })

display(HTML("<b>Table 1: GPC Model Performance</b>"))
display(performance_table)
```

**Table 1: GPC Model Performance**

|   | Metric           | Value             |
|---|------------------|-------------------|
| 0 | Mean AUC-ROC     | 0.775             |
| 1 | SD across folds  | ± 0.046           |
| 2 | Cross-validation | 5-fold stratified |

We fit the model using the full scaled dataset, using the fitted kernel and extracted the optimized length-scales. Once we had the length scales, we could compare each length scale for each feature to one another, showing us which predictive features affected the predictive model more or less (i.e. smaller length scale = higher predictive relevance for CDR).

```
In [ ]: # 6. Fit final model on full dataset to learn optimized kernel hyperparameters
gpcc.fit(X_scaled, y)

# Extract learned ARD Length-scales from the optimized kernel
ls = gpcc.kernel_.get_params()["length_scale"]

feature_names = ["Age", "Educ", "SES", "Sex", "nWBV"]
```

```
In [21]: length_scale_table = pd.DataFrame({
    "Feature": ["Age", "EDUC", "SES", "Sex", "nWBV"],
    "Length-scale": [f"{l:.3f}" for l in ls]
})

# Sort ascending so most influential (smallest) appears at top
length_scale_table = length_scale_table.sort_values("Length-scale").reset_index(drop=True)

# Add interpretation column based on rank
length_scale_table["Interpretation"] = (
    ["Most influential"] +
    [""] * (len(length_scale_table) - 2) +
    ["Least influential"]
)

display(HTML("<b>Table 2: ARD Length-scales by Feature</b>"))
display(length_scale_table)
```

**Table 2: ARD Length-scales by Feature**

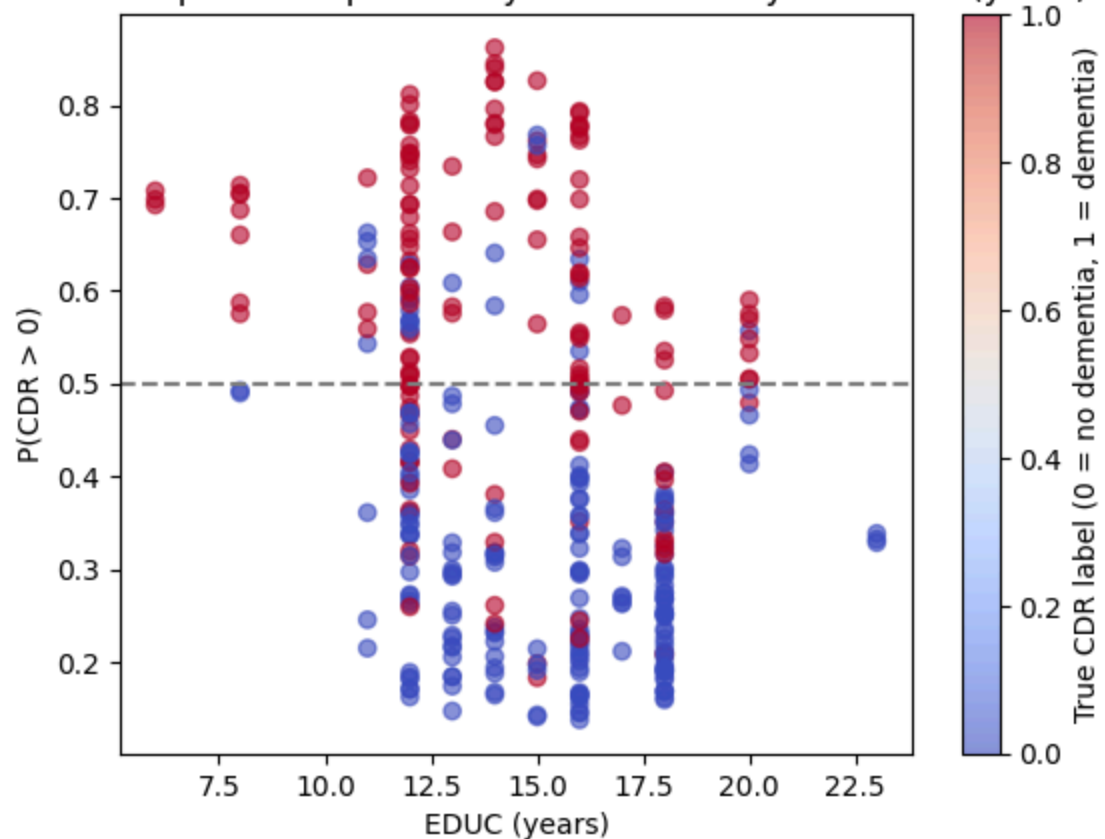
|   | Feature | Length-scale | Interpretation    |
|---|---------|--------------|-------------------|
| 0 | EDUC    | 0.667        | Most influential  |
| 1 | nWBV    | 1.064        |                   |
| 2 | Age     | 1.270        |                   |
| 3 | Sex     | 2.631        |                   |
| 4 | SES     | 2.857        | Least influential |

```
In [29]: import matplotlib.pyplot as plt

probs = gpcc.predict_proba(X_scaled)[: , 1]
plt.scatter(df["EDUC"], probs, c=y, cmap='coolwarm', alpha=0.6)
plt.axhline(0.5, linestyle='--', color='gray')
plt.xlabel("EDUC (years)")
plt.ylabel("P(CDR > 0)")
plt.title("Figure 1. GPC predicted probability of dementia by Education (years)")
plt.colorbar(label="True CDR label (0 = no dementia, 1 = dementia)")
```

Out[29]: <matplotlib.colorbar.Colorbar at 0x7c7596300a40>

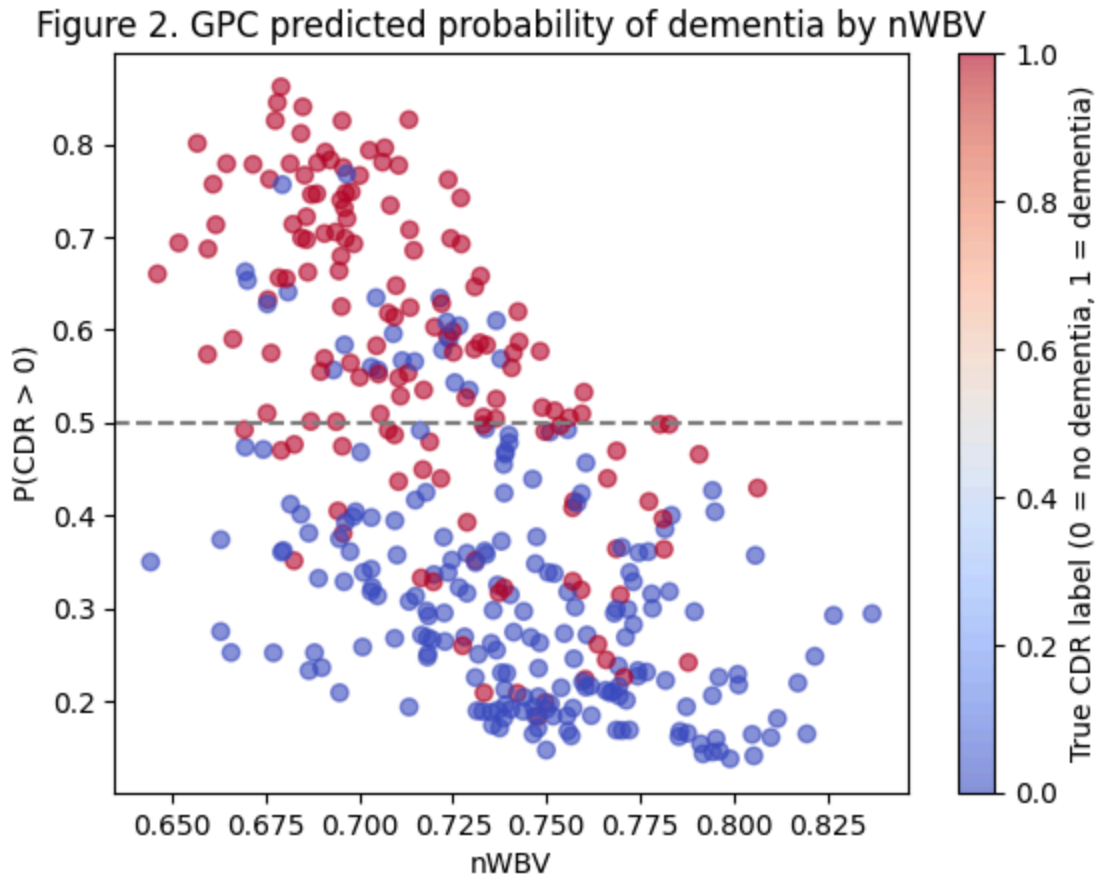
Figure 1. GPC predicted probability of dementia by Education (years)



```
In [30]: probs = gpc.predict_proba(X_scaled)[: , 1]
plt.scatter(df["nWBV"], probs, c=y, cmap='coolwarm', alpha=0.6)
plt.axhline(0.5, linestyle='--', color='gray')
plt.xlabel("nWBV")
plt.ylabel("P(CDR > 0)")
plt.title("Figure 2. GPC predicted probability of dementia by nWBV")
plt.colorbar(label="True CDR label (0 = no dementia, 1 = dementia)")
```

Out[30]: <matplotlib.colorbar.Colorbar at 0x7c7597d12b10>





## Results

Our initial results regarding our model fit were measured via mean AUC score and standard deviation across the cross-validated 5 folds. The mean AUC was 0.775 with a standard deviation of  $\pm 0.046$  (Table 1). Following the model fit, we used the GPC model on our entire dataset to capture the feature-specific lengths scales. These indicated the level of relevance of this feature with respect to variation in the model and relation to CDR. Education had a length-scale of 0.667, nWBV had a length scale of 1.064, Age was at 1.270, Sex was higher at 2.631, and Socioeconomic status was the highest at 2.857 (Table 2). To further examine model behavior, predicted probabilities calculated using the predictive distribution from the posteriors and distribution acquired through Newton's method and the LaPlace approximation, respectively, were plotted against individual features for all patients, colored by true CDR label. Figure 1 displays predicted probability of dementia against EDUC. Despite EDUC returning the smallest length-scale, the plot shows substantial overlap between CDR = 0 and CDR > 0 patients across education levels, with the model expressing considerable uncertainty across the range. Figure 2 displays predicted probability against nWBV, where a clearer gradient is visible — patients with lower nWBV tend to receive higher predicted probabilities of dementia, with red dots (CDR > 0) clustering at lower brain volumes and blue dots (CDR = 0) at higher values. An overlap zone is apparent around nWBV 0.700–0.740, where the model appropriately hedges toward 0.5 rather than committing to a decisive classification.

## Discussion

The GPC model demonstrated moderate discriminative ability with a cross-validated AUC of 0.775 ( $\pm 0.046$ ), suggesting the selected demographic and structural features carry meaningful but incomplete predictive signal for dementia status. We can see that the model is correctly identifying dementia patients, given the features we have included, around 78% of the time. Our AUC may not have been higher due to the exclusion of certain highly correlated features such as MMSE to avoid artificially inflated model performance, as well as the size of the dataset alone. Additionally, since we used demographic and indirect risk factors for measuring probability of dementia ( $CDR > 0$ ), we inherently limit the predictive power where using more direct known predictors like MRI volumetrics and biomarkers would result in better model performance. The relatively low standard deviation across folds suggests the result is stable across different subsets of the data.

The ARD length-scales suggested that education level was the most influential feature in the model, followed by nWBV, age, sex, and SES. This is somewhat at odds with the predicted probability plots, where nWBV showed a clearer visual gradient — a tension worth acknowledging, as length-scales reflect global sensitivity across the entire input space while the plots capture local behavior for individual patients.

A key component of performing GPC is the confidence we have in our results. This is clearly achieved using the cross-validation AUC approach as with other models, but GPC specifically quantifies uncertainty in a unique way. Logistic regression gives a predicted probability for each patient but had equal confidence in each result. GPC produces a predictive variance  $\sigma^2$  along with each predicted probability, reflecting confidence in the estimate. Predictions near 0.5 accompanied by a high  $\sigma^2$  indicate genuine uncertainty (usually regions of that training data that are sparse or the classes overlap) rather than a confident borderline prediction. This can be seen in the Figure 2 predicted probability plot where there is an overlap in nWBV classification around 0.700-0.740, showing that the model hedges rather than force a decision. If this were approached using a logistic regression model, in contrast, every patient at the same nWBV value would get the same predicted probability regardless of how many similar patients were in the training data. In a clinical context this GPC distinction is meaningful - a model that communicates its own uncertainty is more informative than one that produces confident predictions uniformly, particularly for patient profiles that fall outside the range well-represented in the training data. This is especially important because CDR classification is far more nuanced and particular than other disease outcome measures. Logistic regression would say a predicted probability of dementia of 0.61 shows moderate evidence of dementia, not accounting for confidence in this result. GPC would provide a classification like a predicted probability of dementia around 0.54 but say that the patient is in an ambiguous zone and more information would be needed to make a clinical judgement regarding cognitive impairment. Clinically, this can mean significantly different things with respect to diagnosis and screening.

The uncertainty estimates that make GPC clinically valuable — the  $\sigma^2$  values pulling ambiguous predictions toward 0.5 — are themselves a product of the Laplace approximation. Because the true posterior over latent risk scores is intractable, the Laplace approximation fits a Gaussian at the mode of the posterior to characterize uncertainty around that peak. So, the confidence the model expresses in any prediction is only as reliable as Gaussian approximation is to the true posterior. In well-observed regions of the input space this tends to hold reasonably well, but in sparse regions — exactly where honest uncertainty matters most clinically — the true posterior may deviate from the fitted Gaussian, meaning the uncertainty estimates themselves carry some error. We could use more concrete and complex inference methods here such as Markov Chain Monte Carlo or expectation propagation, but there would be a much higher computational cost.

## References

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